

IT1244

Artificial Intelligence: Technology & Impact

Week 9 Tutorial 7

TODAY'S AGENDA

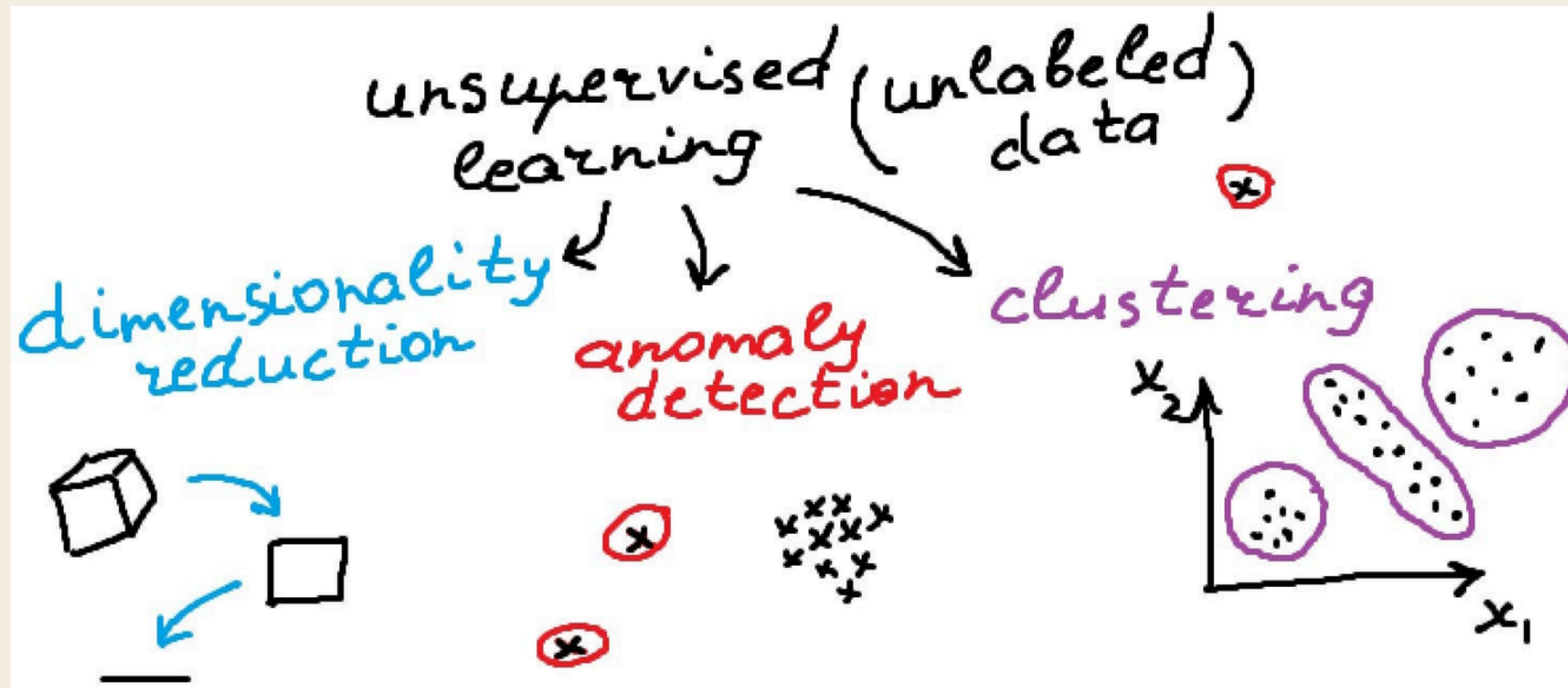
1. Recap of concepts
2. Discussion of tutorial questions
3. Pollev Questions



RECAP

UNSUPERVISED LEARNING

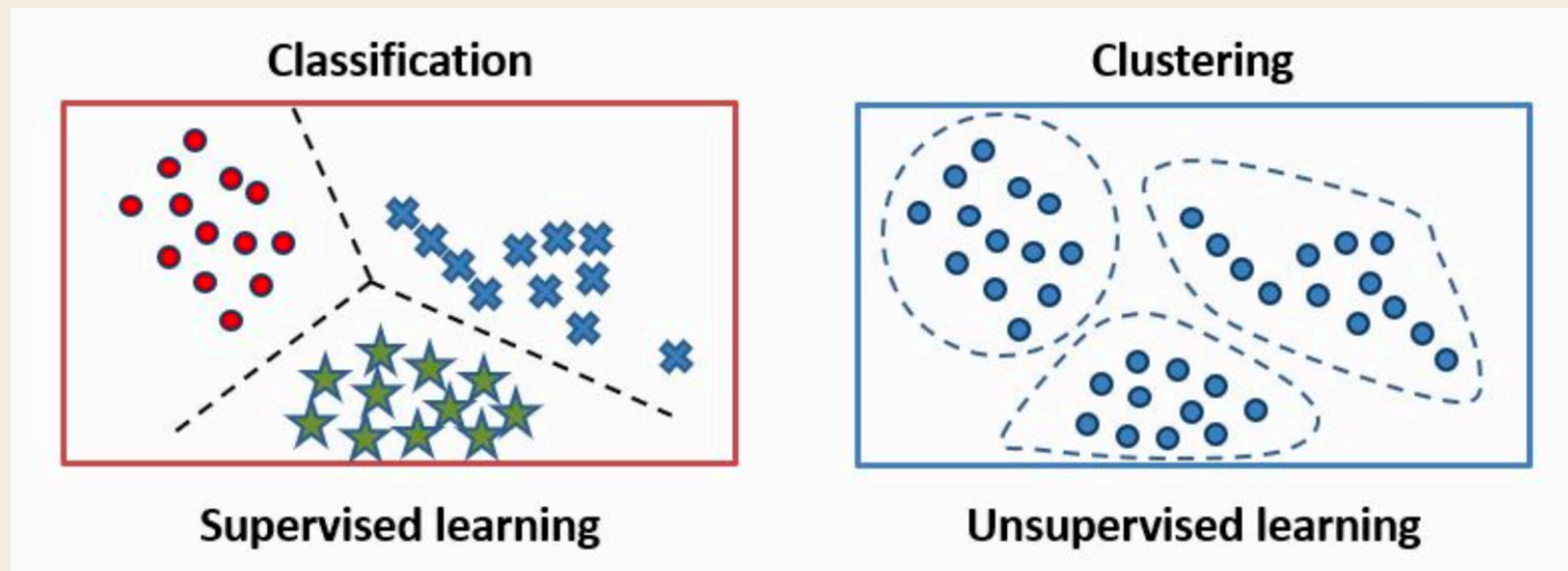
Algorithm that learns without labelled data (no output labels given!)



Useful when labeled data is unavailable or too costly (which is very common irl!)

CLUSTERING

- Clustering finds natural groupings in data based on similarity
 - Even when we don't know the labels
- Used in: Customer segmentation, image compression etc



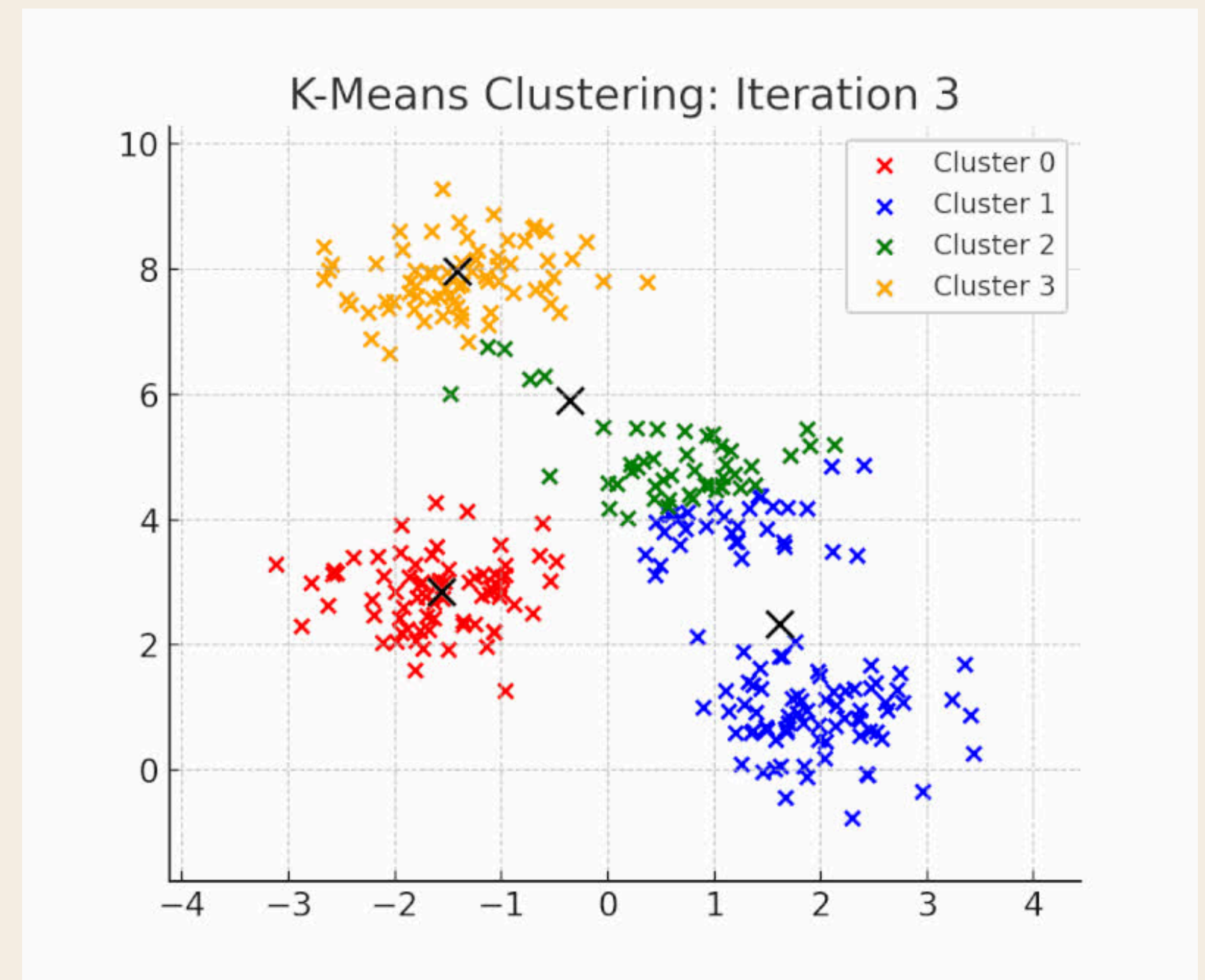
⚠ Clustering doesn't tell you what each group is - it just groups similar points together. You'll need to decide if those groupings are useful or interpretable.

K-MEANS CLUSTERING

Goal: Partition data into K clusters by minimizing the distance between points and their assigned centroid.

How It Works:

1. Initialize K centroids (randomly)
 2. Assign each data point to its **closest** centroid
 3. Move each centroid to the **mean** of its assigned points
 4. Repeat Steps 2-3 until centroids no longer change or maximum iterations is reached.
- ! “closest” is usually measured using Euclidean distance



K-MEANS CLUSTERING

Qn: Do we need to do feature scaling for K-Means?

Yes. K-means relies on **distance calculations**, so features with larger ranges can dominate the distance measure if the data is not scaled

Qn: Do we always initialise centroids from one of the samples?

No. While usual convention (scikit-learn) is to initialise some points as the centroids, you can define where you want to start your centroids

Qn: So the selected points to be centroid will move around during K-means?

No. The data points remain fixed. The centroid starts as a copy of a selected data point, and this centroid moves to the mean of its assigned cluster.

WITHIN-CLUSTER SUM OF SQUARES

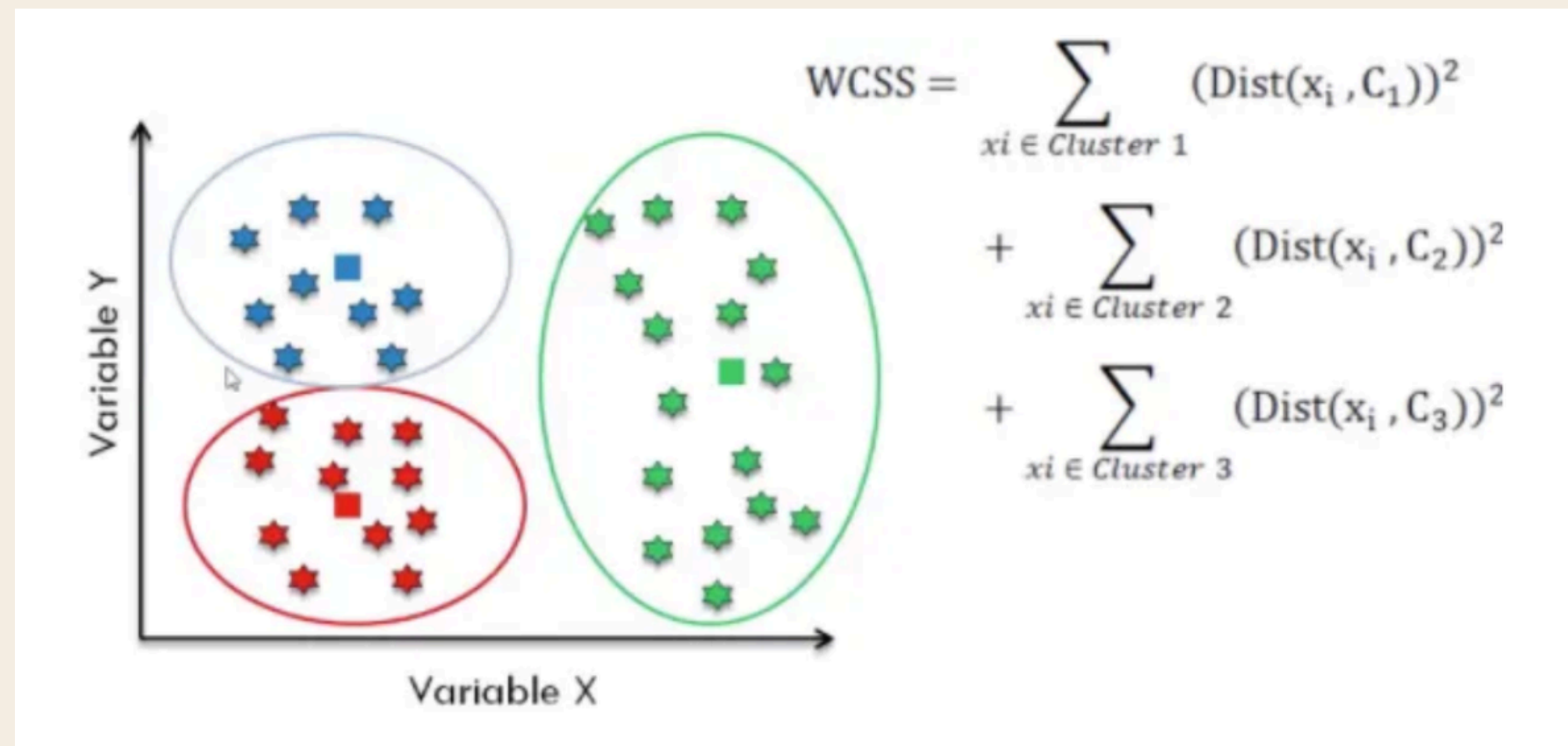
Within-Cluster Sum of Squares (WCSS) is the sum of squared Euclidean distance between each sample and the centroid of its cluster.

$$WCSS = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

C_i = Cluster i

μ_i = Centroid of Cluster i

x = Data point of Cluster i



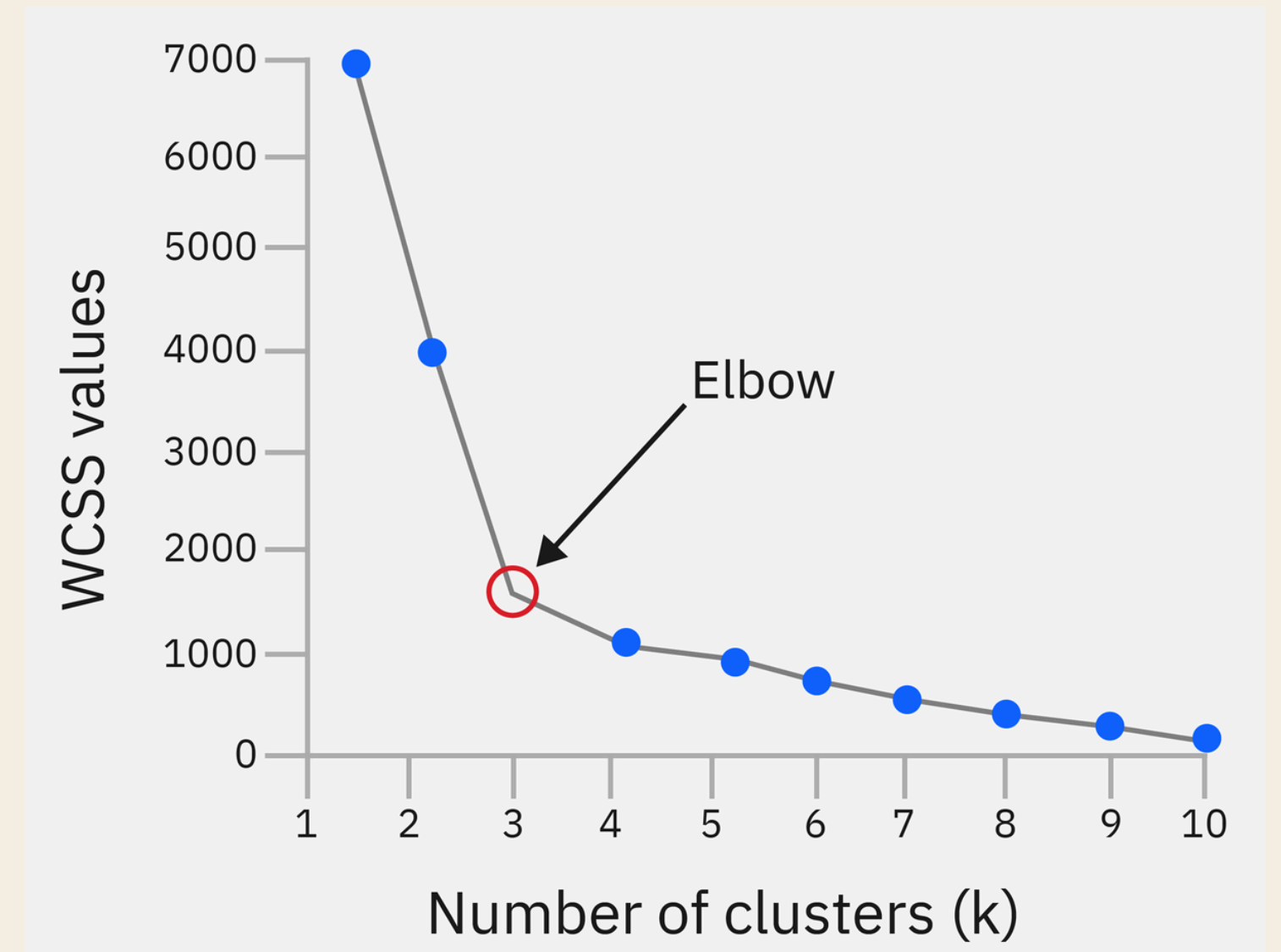
HOW TO CHOOSE K?

1. Elbow Method

- Compute WCSS for different values of K
- Choose K where the decrease in WCSS starts to slow down

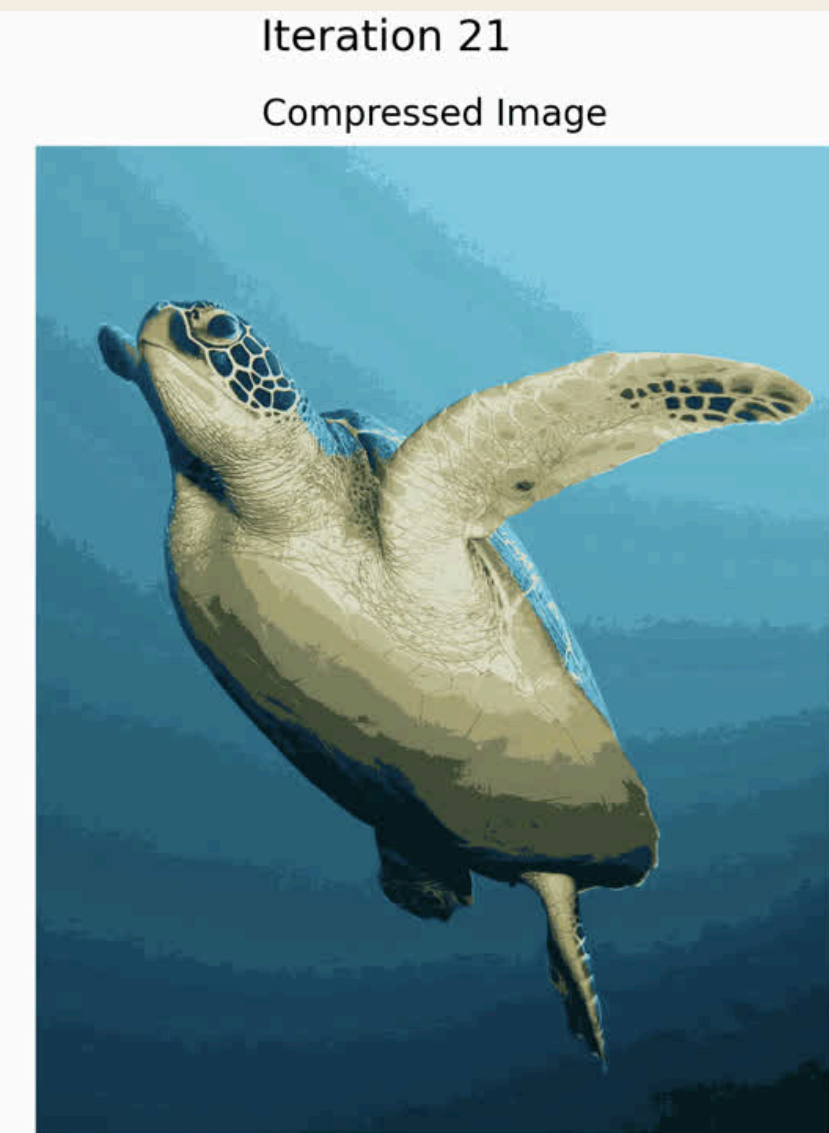
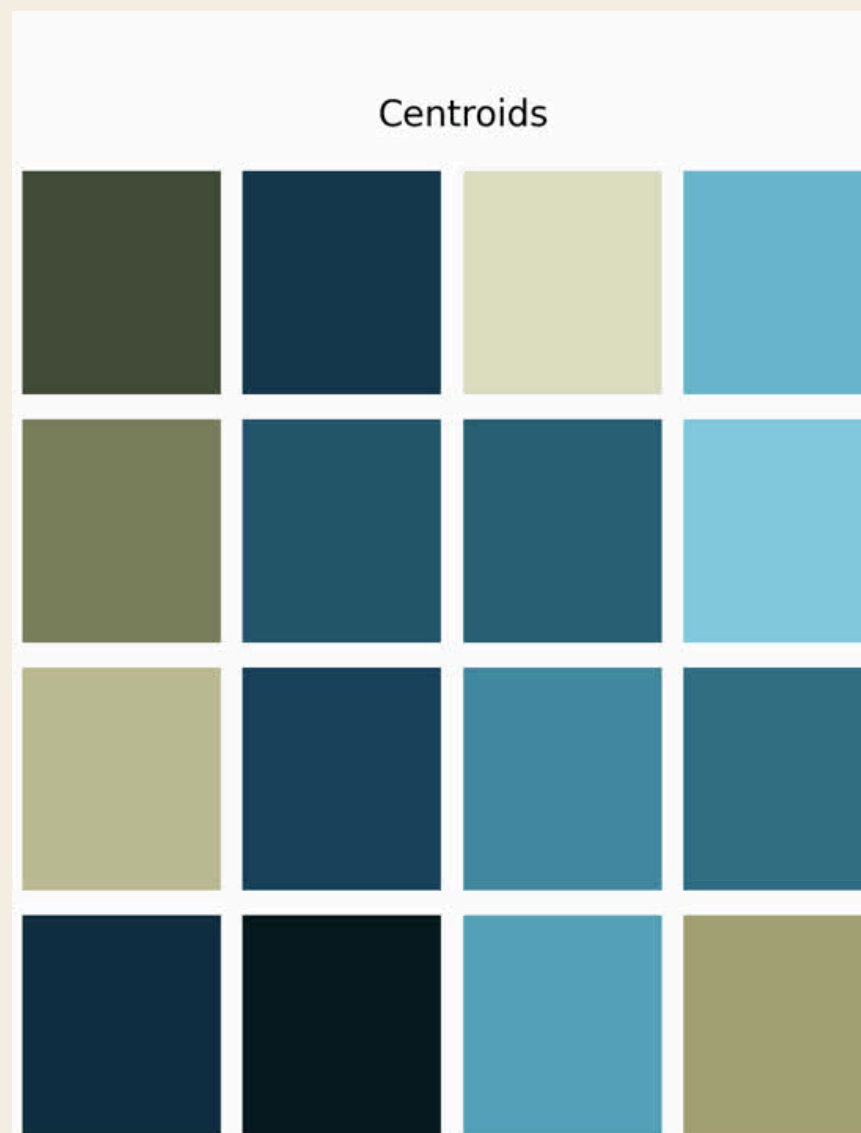
⚠ Increasing K always reduces WCSS, but too large K may produce meaningless clusters (Q5)

2. Choose K manually (based on domain knowledge)



CLUSTERING IN IMAGE COMPRESSION

- K-Means clustering groups similar colors (RGB values) together into k clusters.
- Instead of storing the full RGB value for each pixel, we:
 - Replace each pixel's color with the nearest cluster center (centroid)
 - Only store the k color centroids + the pixel's cluster label.



TUTORIAL QUESTIONS

TUTORIAL QUESTION 1A

Use the K-means clustering algorithm with $K = 2$ to cluster the data.

(a) Using the initial cluster assignments provided, compute the initial centroids of the two clusters.

X1	X2	Cluster-ID
5	0	1
5	2	2
3	1	1
0	4	2
2	1	1
4	2	2
2	2	1
2	3	2
1	3	1
5	4	2

	1	2
X1-Centroid	2.6	3.2
X2-Centroid	1.4	3

TUTORIAL QUESTION 1B

(b) Reassign each data point to the nearest centroid using Euclidean distance, and compute the new cluster assignments and updated centroids.

			after 1st iter
Distance	1	2	new Cluster id
	7.72	12.24	1
	6.12	4.24	2
	0.32	4.04	1
	13.52	11.24	2
	0.52	5.44	1
	2.32	1.64	2
	0.72	2.44	1
	2.92	1.44	2
	5.12	4.84	2
	12.52	4.24	2

X1	X2	Cluster-ID
5	0	1
5	2	2
3	1	1
0	4	2
2	1	1
4	2	2
2	2	1
2	3	2
1	3	2
5	4	2

	1	2
X1-Centroid	3	2.833333
X2-Centroid	1	3

TUTORIAL QUESTION 1C, 1D

- (c) Repeat the K-means update process until 3 iterations are completed.
- (d) Report: the cluster centroids after each iteration, the final cluster assignment of each data point after the third iteration.

	1	2
X1-Centroid	2.6	3.2
X2-Centroid	1.4	3

	1	2
X1-Centroid	3	2.83333333
X2-Centroid	1	3

	1	2
X1-Centroid	3.8	2
X2-Centroid	1.2	3.2

	1	2
X1-Centroid	4	1.25
X2-Centroid	1.6666667	3

			after 3rd iter
Distance	1	2	new Cluster id
	2.88	19.24	1
	2.08	10.44	1
	0.68	5.84	1
	22.28	4.64	2
	3.28	4.84	1
	0.68	5.44	1
	3.88	1.44	2
	6.48	0.04	2
	11.08	1.04	2
	9.28	9.64	1

TUTORIAL QUESTION 2A

K-means++ is an improved version of the K-means clustering algorithm that focuses on better initialization of cluster centroids. What is the key difference between K-means and K-means++?

K-means: picks the initial centroids completely randomly.

- If you get unlucky and two centroids start close together, they'll both compete for the same region and you end up with a bad final clustering.

K-means++: picks the first centroid randomly, but each subsequent centroid is chosen with **probability proportional to its distance from the chosen centroids.**

- So points that are far away from existing centroids are more likely to be picked next.

TUTORIAL QUESTION 2B

(b) How does the K-means++ initialization work

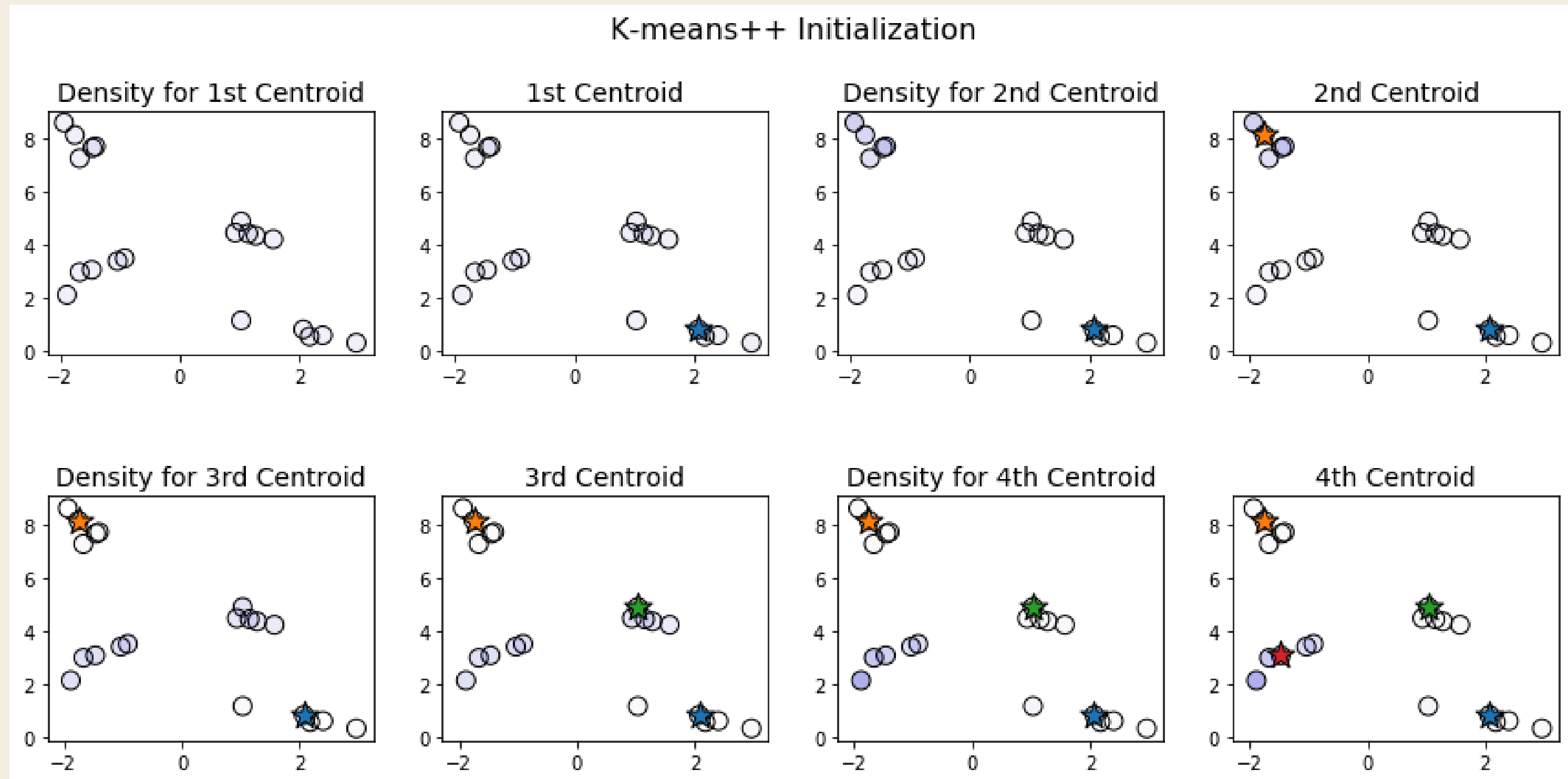
The K-means++ centroid initialization process works as follows:

1. Randomly select the first centroid from the dataset.
2. Compute the distance between each data point and the nearest existing centroid.
3. Select the next centroid with probability proportional to the square of this distance. **(Note: It's still random! Just not uniformly chosen anymore)**
4. Repeat Steps 2-3 until K centroids have been chosen.

After initialization, the standard K-means clustering algorithm proceeds normally.

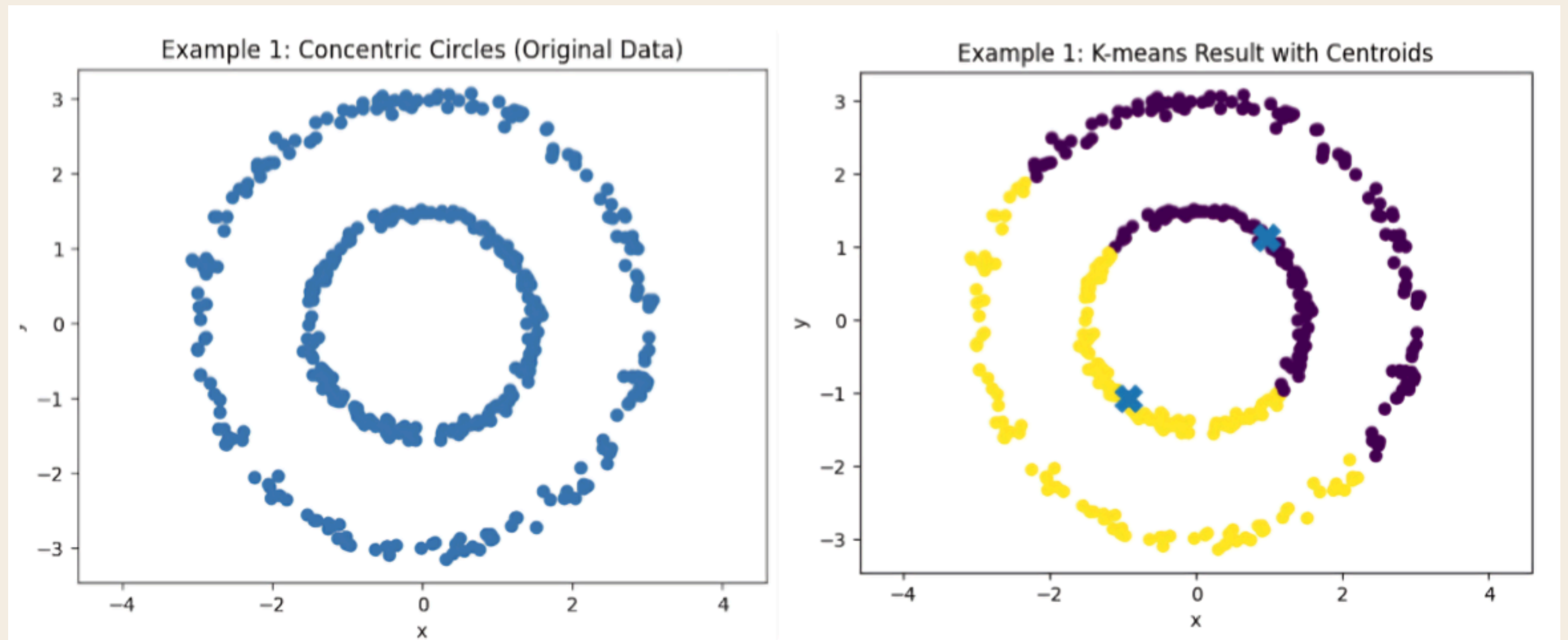
TUTORIAL QUESTION 2B

(b) How does the K-means++ initialization work



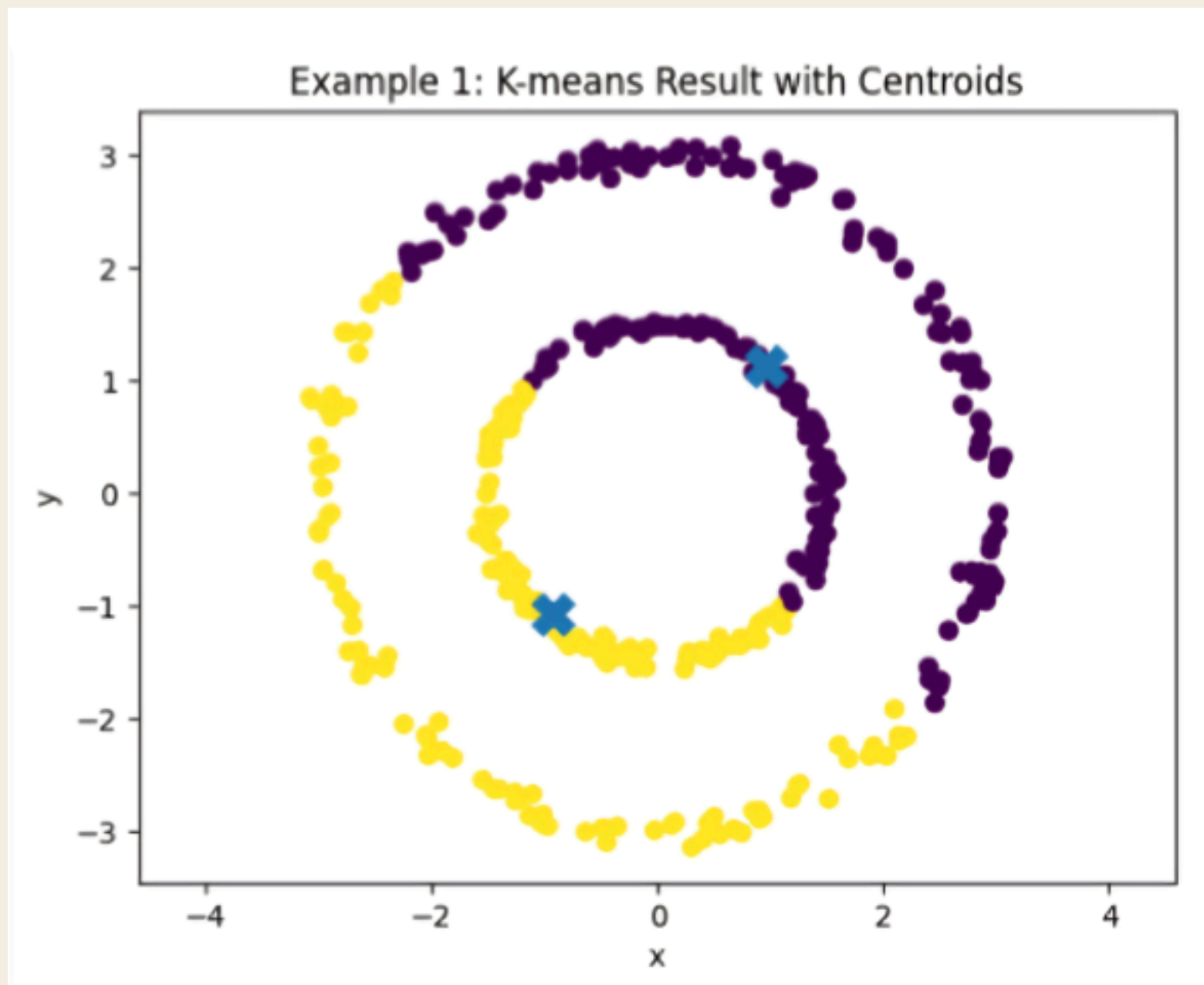
TUTORIAL QUESTION 3

Draw an example dataset in 2D space where the K-means algorithm with $K = 2$ fails to produce meaningful clusters.



TUTORIAL QUESTION 3

Draw an example dataset in 2D space where the K-means algorithm with $K = 2$ fails to produce meaningful clusters.



- K-means assigns each point to its nearest centroid. That's the only rule it follows.
- This means cluster boundaries are always **straight lines**. No matter how you place 2 centroids, you can never produce a circular boundary (so separating an inner ring from an outer ring is geometrically impossible for K-means)

TUTORIAL QUESTION 4

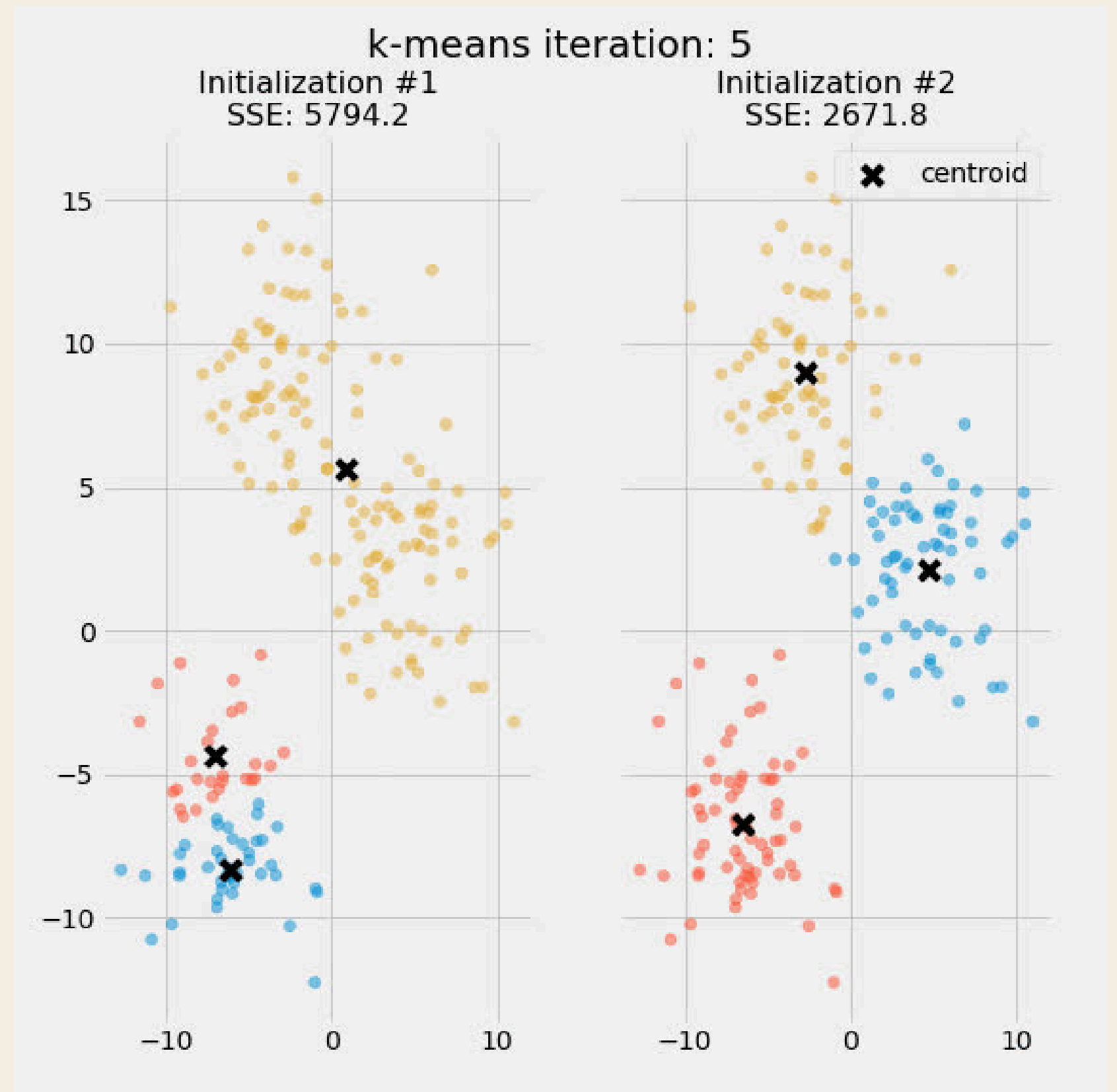
It is common to run the K-means algorithm multiple times for the same value of K , since the results depend on the initial placement of centroids.

(a) How can K-means be run multiple times to obtain different clustering results?

- By initializing the cluster centroids with different random starting points.
- Each run may produce a different clustering result because the algorithm may converge to different local minima.

TUTORIAL QUESTION 4

K-means is highly sensitive to initial placement of centroids!!



TUTORIAL QUESTION 4

In Code:

```
from sklearn.cluster import KMeans  
  
kmeans = KMeans(  
    n_clusters=3,  
    n_init=20, # run K-means 20 times  
    random_state=None  
)  
  
kmeans.fit(X)
```

After the 20 runs,
only the “best”
clustering is returned

🤔🤔 But how do we
define “best”? Next
question asks this!

TUTORIAL QUESTION 4

Q: How do we choose the best result among multiple runs?

Select the run that minimizes the K-means objective function, i.e, the within-cluster sum of squares (WCSS)/ Inertia.

$$WCSS = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

C_i = Cluster i

μ_i = Centroid of Cluster i

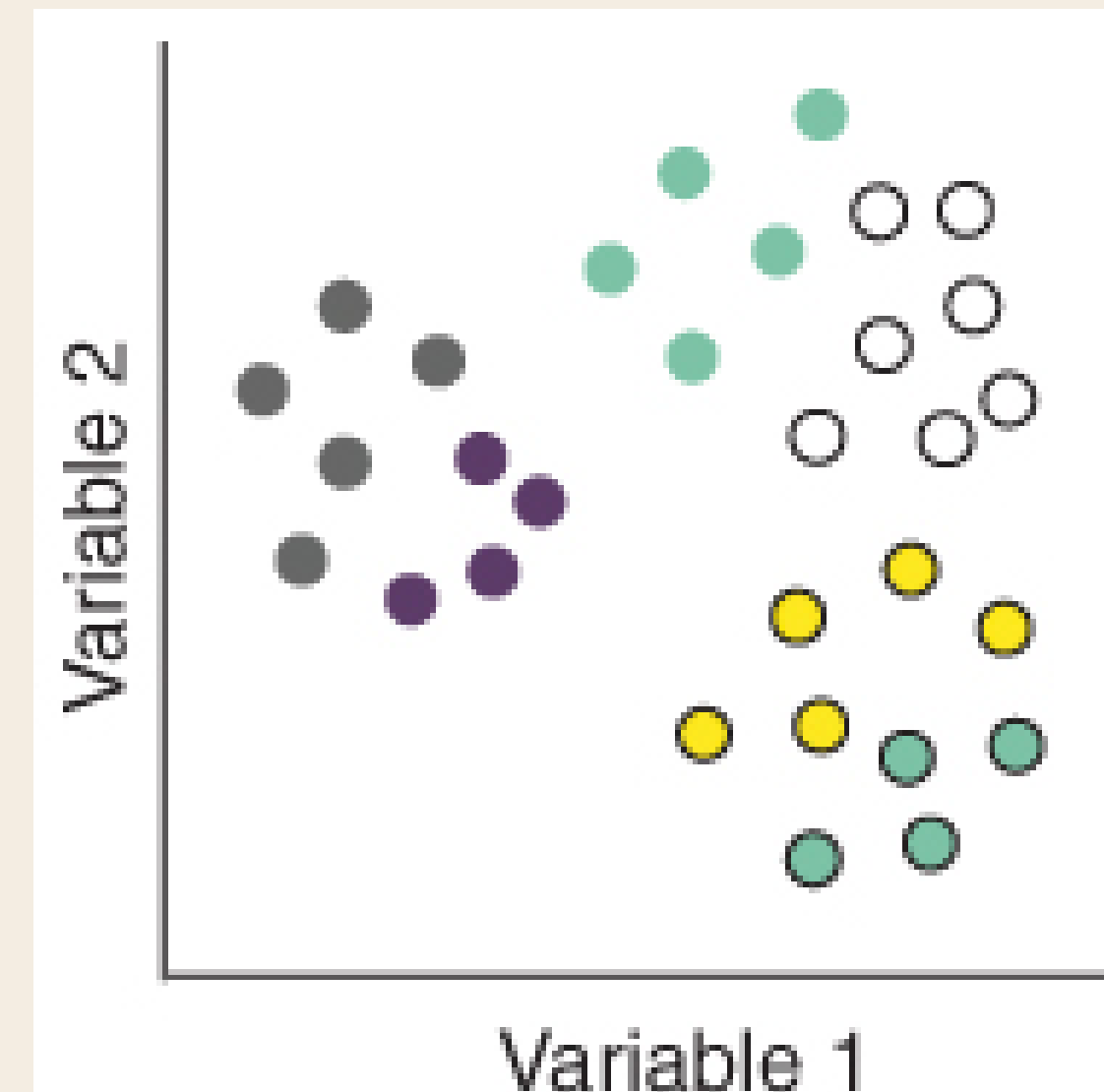
x = Data point of Cluster i

Idea: Lower inertia indicates that the data points are closer to their assigned centroids, meaning more compact clusters.

TUTORIAL QUESTION 5

Q: Is overfitting possible in unsupervised learning (eg. in K-means)?

- Yes definitely possible!
- Too many clusters ($K = N$): each point becomes its own cluster $\rightarrow$ within-cluster variance = 0
- The algorithm perfectly “fits” the dataset but learns no meaningful structure.



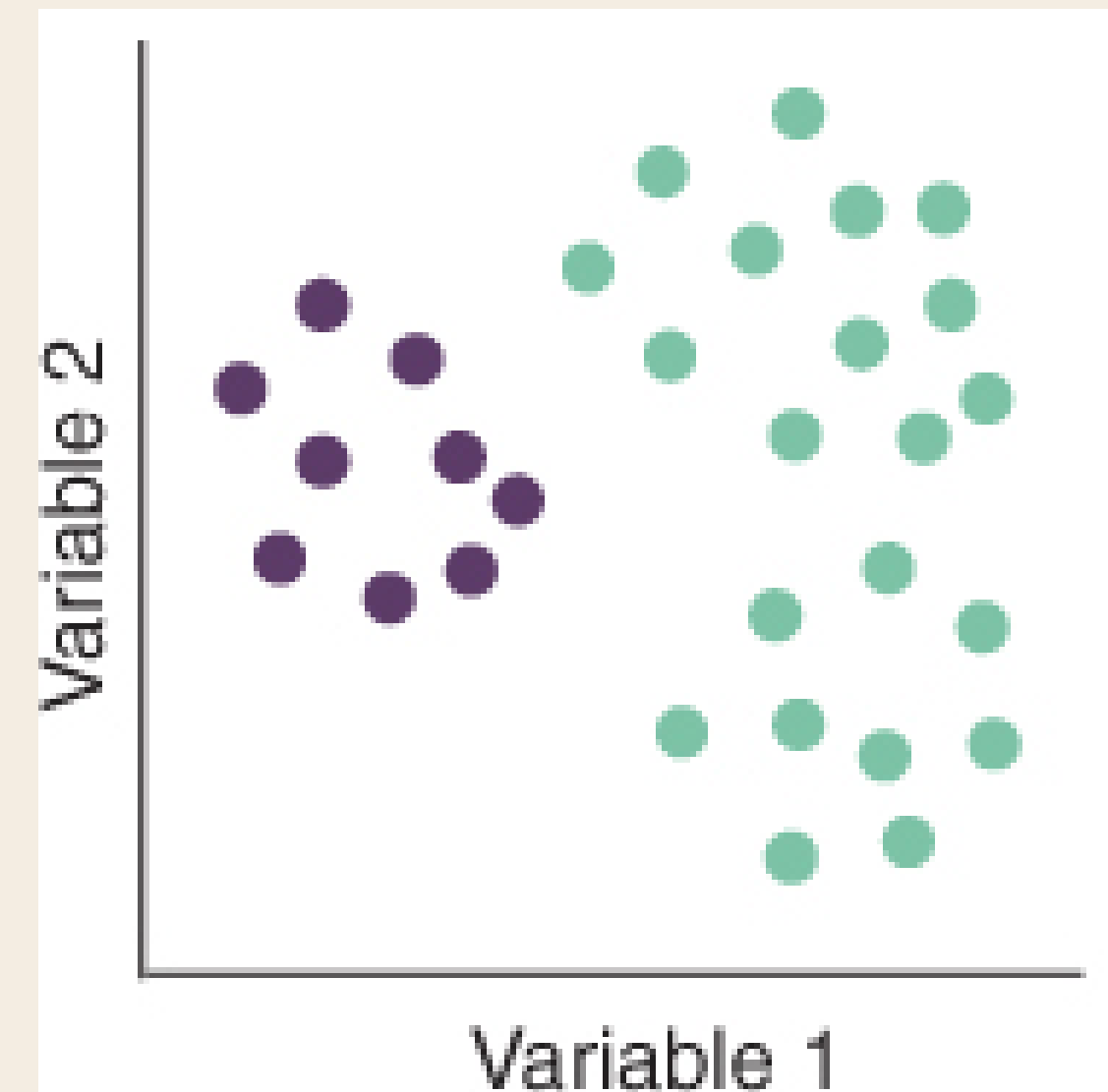
OVERFITTING

TUTORIAL QUESTION 5

Q: Is overfitting possible in unsupervised learning (eg. in K-means)?

- In fact, underfitting is also possible!
- Too few clusters (eg. $K = 1$)
- All points in one cluster $\rightarrow$ true groups are ignored

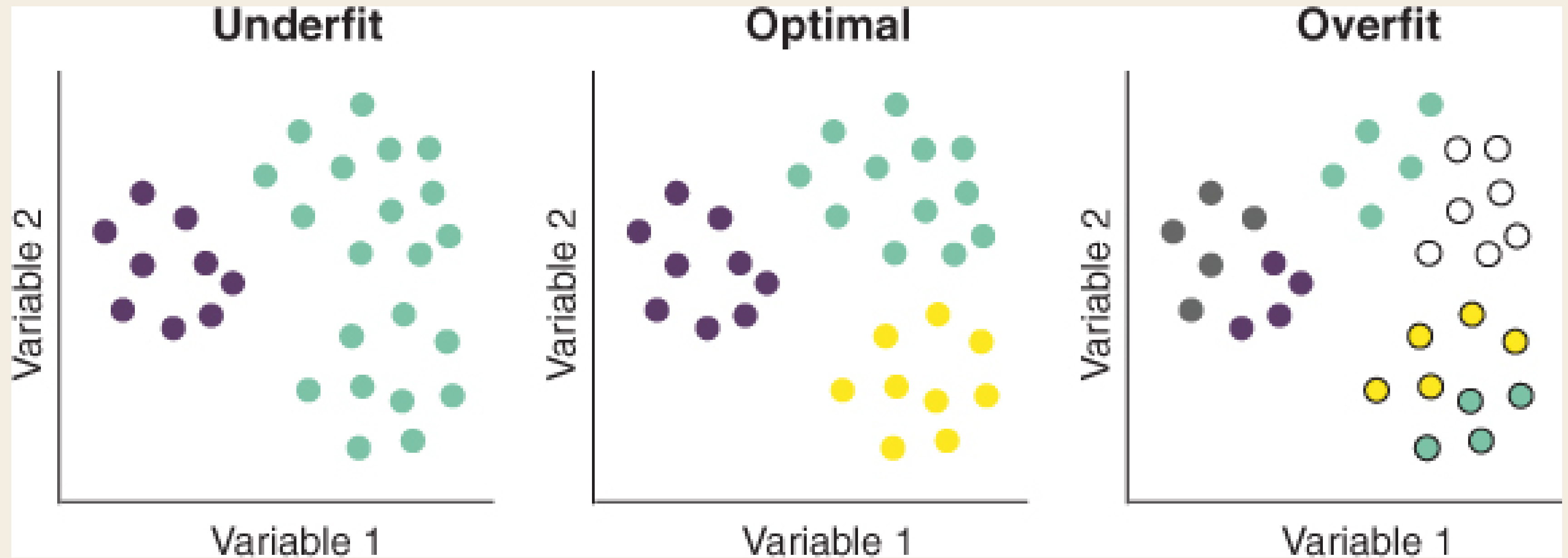
Selecting an appropriate value of K is important to balance between overfitting and underfitting in clustering.



UNDERFITTING

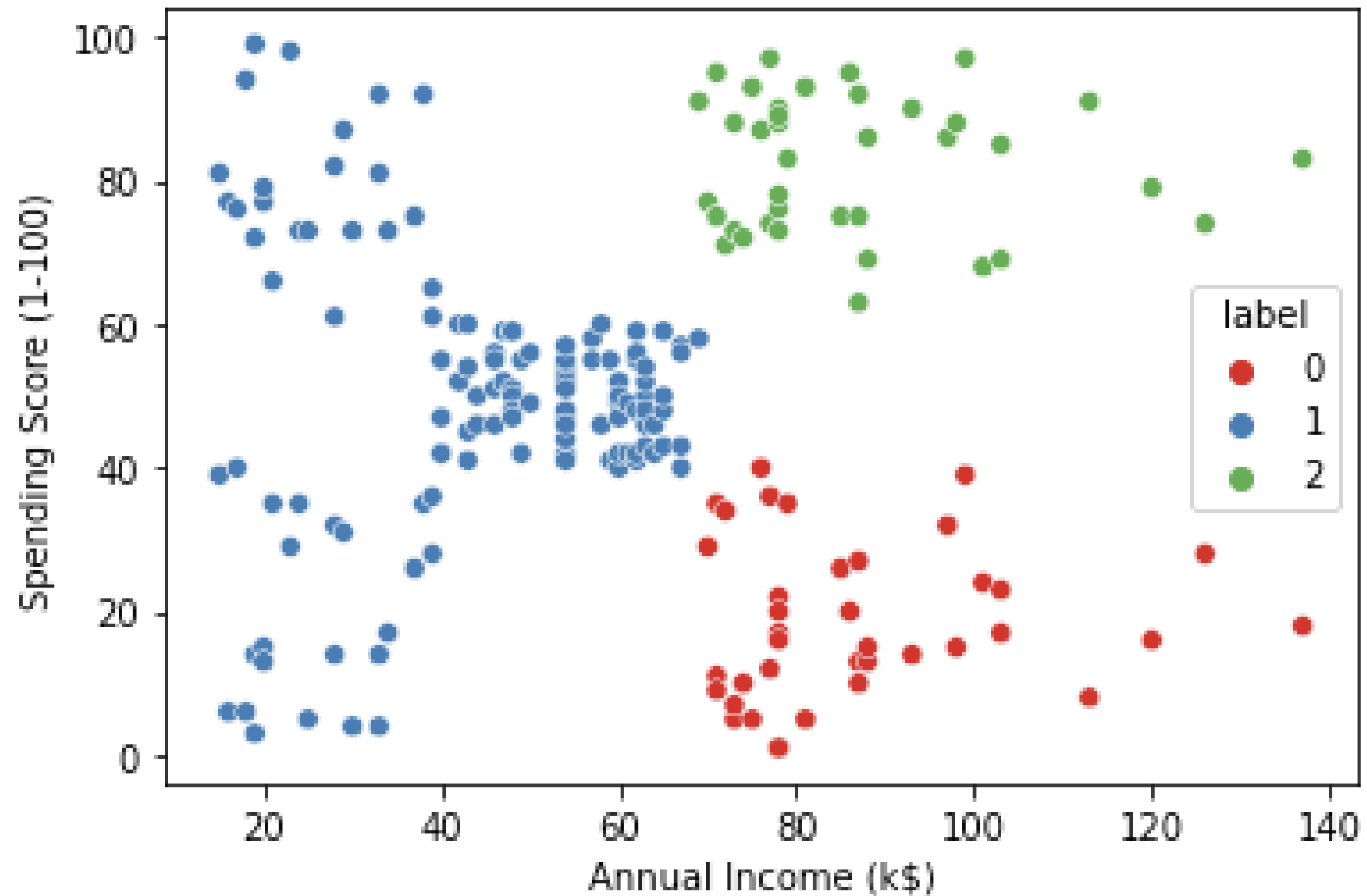
TUTORIAL QUESTION 5

Q: Is overfitting possible in unsupervised learning (eg. in K-means)?



TUTORIAL QUESTION 5

What do you think: Is this overfitting / underfitting?



POLLEV QUESTIONS

Join by Web:

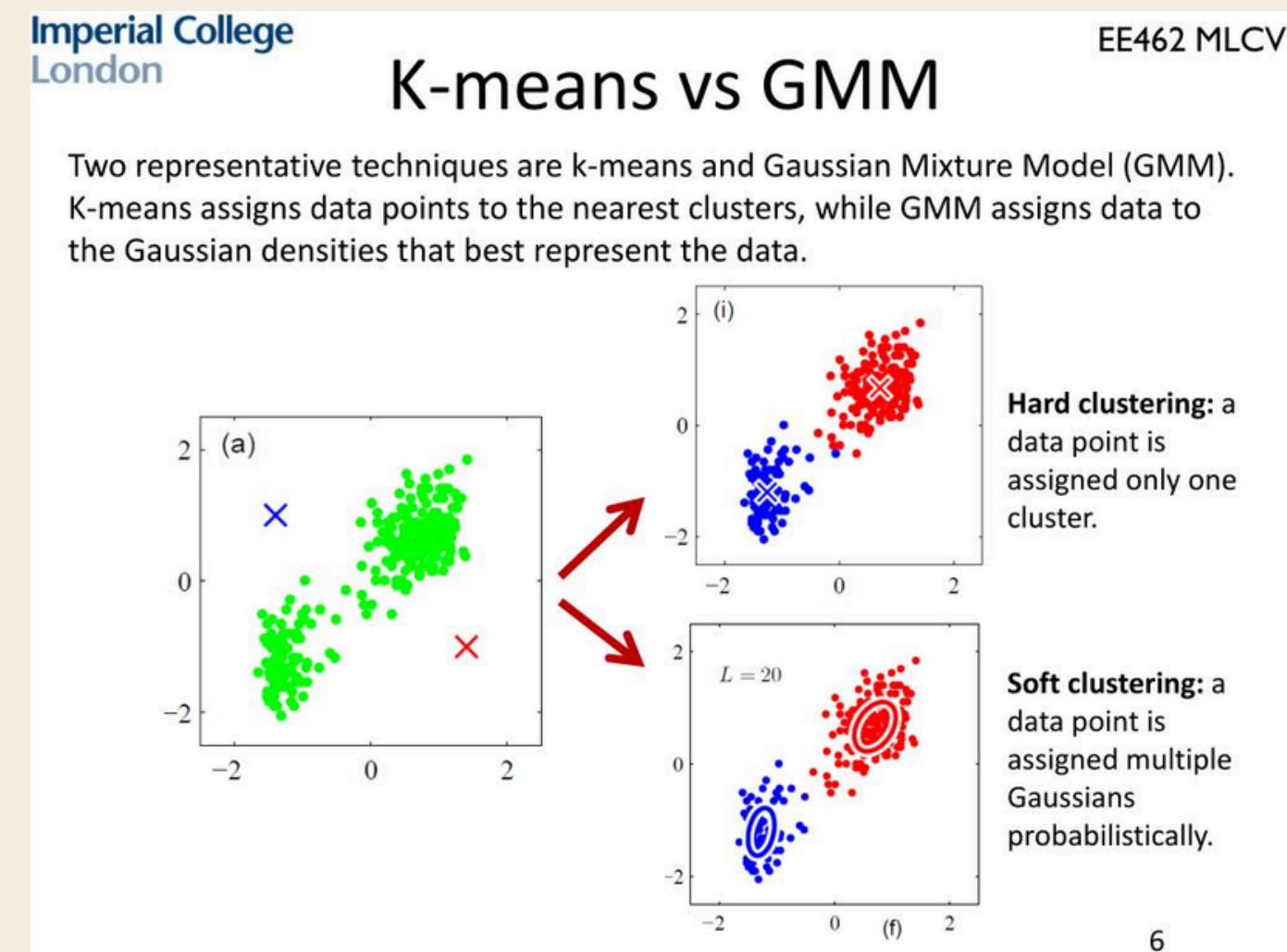
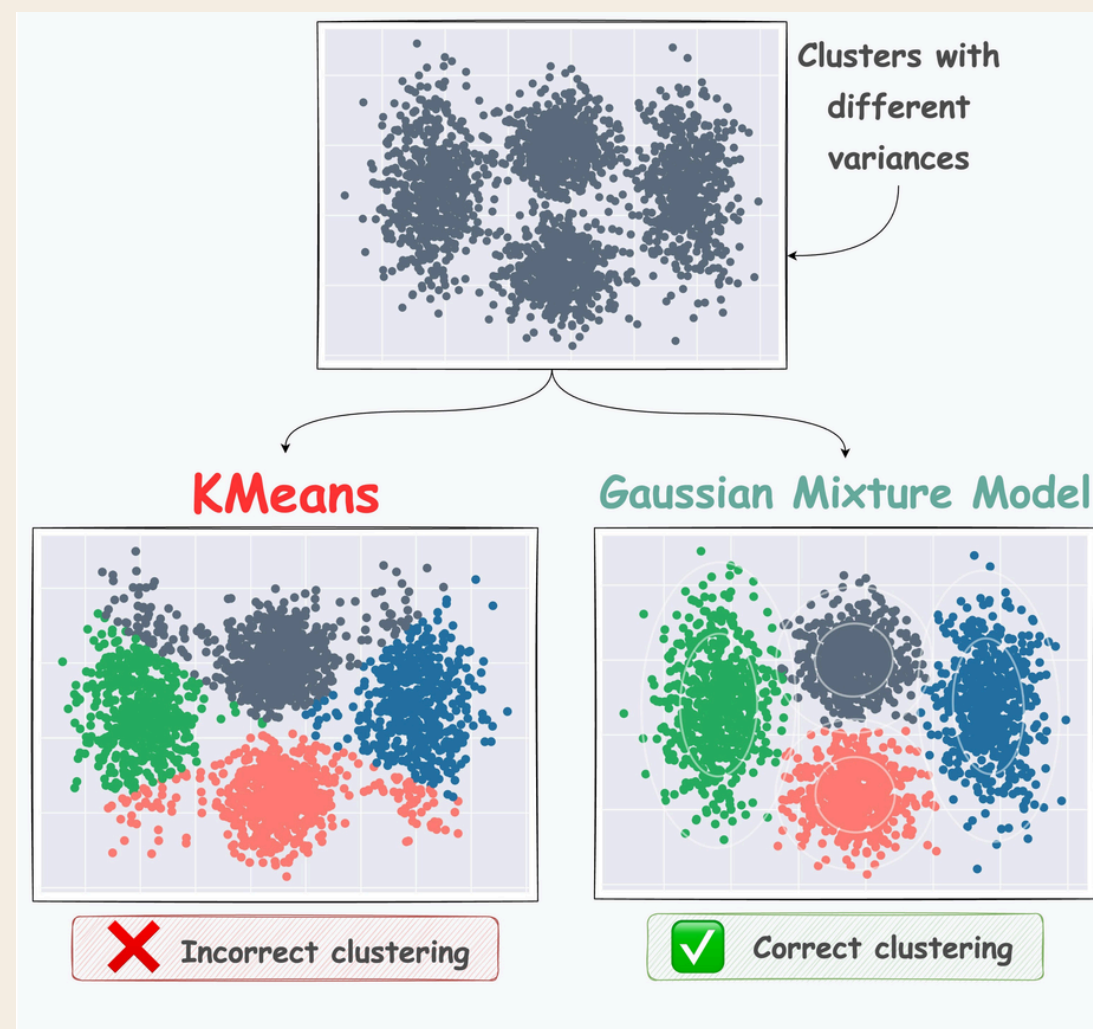
PollEv.com/kaironglee

Join by QR:



EXTRA READINGS

- Another type of Clustering: Gaussian Mixture Models
- Intro to Dimensionality Reduction
- Deeper into Dimensionality Reduction: t-SNE, UMAP



NEXT WEEK

Neural Network

People with no idea
about AI, telling me my
AI will destroy the world



Me wondering why my
neural network is
classifying a cat as a dog..

